POSHA

ELECTRONICS ASSIGNMENT – KABIR KRISHNA – 23/EE/120

TASK1:-

A load cell is a transducer that converts applied force into a proportional electrical signal using strain gauges arranged in a Wheatstone bridge.

LOAD Cell Objective:-

1. Load is applied at **Pan + Stirrer**
2. Robotic arm picks from:
   Macro containers (bulk ingredients)
   Spice rack (micro ingredients)
3. Disturbances present:
   Stirring motion/Noise Disturbances
   Arm movement
   Heat from oven+cooking
   Dislocated Pressure on Pan
   Envirnomental Factors(Oil+Steam)

Load Cell Placement:-

- Best position is **Under the pan (Pan + Stirrer base)**
- **CAUSE:-**

1. Measures **actual dispensed food**
2. Independent of robotic arm errors
3. More accurate for cooking

Type Of Load Cell Required:-

- **High Precision Single-point load cell (platform type)** made of Stainless Steel with Hermetic Sealing.
  Load Cell Model :- To handle the heat, moisture, and shifting pressure , the best professional-grade series : **HBM PW15AH (Stainless Steel)**
  This is a "**hermetically sealed**" IP68/IP69K sensor. It is designed for food industry "wash-down" environments and can withstand the steam and heat of an oven much better than standard aluminum cells.

Why perfect here:

- Pan is like a **platform load**
- Essential for **Dislocated Pressure** as it handles **off-center loads** (food won't be uniform)
- Compact for integration below stove plate

<u>Load Cell Specifications</u>: Needed to handle the Mechnanical , Electrical , heating and other disturbances specified above and formulated the Load Cell specs accordingly.

1. **Load Cell Capacity** : In robotics and automated food processing, a **safety factor of 2x** is standard. As when the robotic arm drops 1.5kg of ingredients(Payload Mentioned) , the **impact force** (dynamic load) can momentarily double the perceived weight.
   Stirring Torque: If your stirrer pushes against the side or bottom of the pan, it adds "false" weight that can max out a smaller sensor.
   Load Cell Capacity = 2*((weight of Pan +stirrer) + weight of Payload)
   = 2*(1Kgs+1.5Kgs) = 5**Kgs**
   (Assumed weight of Pan + stirrer = 1Kgs)

   Ideal Choice: A **5kg capacity** load cell provides the best balance between high resolution (sensitivity to small ingredient additions) and enough "Tolerance" to prevent permanent deformation from impact.

2. **Thermal Protection :** Must include **PTFE (Teflon) or Glass-filled PTFE thermal spacers** between the pan and the load cell to isolate/Shield the sensor from the 26cm tri-ply pan's heat.
   Thermal Heating: Tri-ply pans reach **200°C–250°C**, which can destroy standard load cells (limited to **60-80°C**).
   PTFE Strengths: It acts as an excellent **thermal barrier** and remains stable up to **260°C** without melting.
   The "Creep" Risk: Under heat and pressure, PTFE **deforms (squishes)**, causing false weight readings and sensor drift.
   Material Upgrade: Using **Glass-filled PTFE** provides better structural rigidity to prevent this deformation.

3. **Handle Oil/Moisture Disturbance**:- are mitigated using IP-rated sealed load cells, protective enclosures with gaskets, conformal coating on PCBs, and shielded cables with proper sealing to prevent ingress and signal noise.
   IP-Rating required – IP65+ as it is Dust tight & Protected from oil splashes & water jets

4. **Accuracy**: Load cell accuracy is categorized by OIML R60 classes. For an industrial food-robotics application, we would prefer:

   Class C3 (Standard): Most commonly used for industrial weighing and dosing. It offers a combined error (linearity, hysteresis, and temperature effect) of approximately **±0.02% to ±0.03% of Full Scale Output (FSO)**.

Class C6 was also an option as it is highly precised but it lags in this system as C3 has leverage as:-

1. Robustness Against "Real-World" Noise:- Any noise from stirrer or the Robotic ARM a C3 cell is physically more "forgiving" and less likely to report ghost weights from the structural flex of the range top.

2. **Sufficient Precision for Cooking:-**
   Let's look at the math for your **5kg capacity** C3 load cell:
   C3 Resolution: 3,000 divisions.
   Calculation: 5000g/3000 = 1.6 grams.

In Practice: Most culinary recipes (even professional ones) don't require precision finer than 1 or 2 grams.

3.**Economic and Thermal Stability:-**
   Cost: C3 cells are significantly cheaper and more widely available in the Stainless Steel / IP65 formats we need for food safety.

5. **Sensitivity: 2.0 mV/V**
   Sensitivity(or Rated Output) defines how much electrical signal the load cell produces per volt of excitation at its maximum capacity.
   Preferred Value: 2.0 mV/V ± 0.1% is the industry standard for high-precision strain gauge load cells.
   As **2.0 mV/V** is more common and offers better stability in the **high-heat environment** of a smart oven.
   Zero Balance: Zero balance of **±1.0% R.O is preferred.** This indicates the sensor's ability to return to a true zero after the robotic arm removes a heavy container.

TASK2:-

ADC Specifications:-For the **Posha** autonomous cooking setup, the best ADC for precise measurement and easy microcontroller integration is the **Texas Instruments ADS1232**.

Specifications of ADS1232:-

- **Integrated Low-Noise Amplifier:** It features a built-in **Programmable Gain Amplifier (PGA)** (essentially an **instrumentation Amplifier**) with selectable gains (1, 2, 64, or 128). For our load cell, setting the gain to **128** allows the system to resolve the tiny millivolt signals from the sensor without needing an external amplifier.
- **High Resolution:** With **24-bit resolution** , it can detect minute weight changes critical for precisely dosing spices from the **Micro Subsystem**.
- **Noise Rejection:** It has a dedicated 10 Samples Per Second (SPS) mode that provides **simultaneous 50Hz and 60Hz rejection**. This is essential for filtering out electrical interference from the oven's heating elements and mechanical noise from the **Stirrer Subsystem**.

- **Environmental Monitoring:** It aslso includes an **internal temperature sensor**, which can be used to calibrate and compensate for weight drift caused by the heat from the 26cm tri-ply pan.
- **Simple Interface:** It uses a **pin-driven control system** with no internal registers to program. This significantly simplifies our embedded code, as we only need to "bit-bang" or use a standard serial interface to pull data into our MCU

While the **HX711** was also a **popular** choice for a chip while selecting the **Right ADC** for our system but the **ADS1232** is an **industrial-grade solution** that offers significantly **lower noise** (**17nV** vs. ~50nV) and superior **50Hz/60Hz power-line rejection**, which is critical near high-power oven elements.
Ultimately, the ADS1232 provides the **high-fidelity signal** integrity required for **robotic dosing** in a mechanically "**noisy**" kitchen environment where an HX711 would suffer from frequent **data jitter**.

TASK 3:-

CABLES & CONNECTORS USED:-

To ensure the **Posha(Smart Oven)** system we must maintain high accuracy while the robotic arm is moving and the 26cm pan is heating, we must use **industrial-grade cabling** and **connectors** that provide **Electromagnetic Interference (EMI) shielding** and **vibration resistance**.

1. Cable Selection: 4-Core Shielded Twisted Pair (STP)

Standard unshielded wires will act as antennas, picking up electrical "noise" from the stirrer motor and the oven's induction/heating elements.

- **Specifications: 24 AWG, 4-Core Shielded Twisted Pair**.
- **Twisted Pair** used cause Differential signals (Signal+ and Signal-) twisted together naturally cancel out electromagnetic interference.
- **Shielded cable** provides an **outer braided copper or foil shield**, when grounded to the MCU's chassis ground, drains away external noise before it reaches the sensitive millivolt signal from the load cell.
- **Jacket Material:** Used **PTFE (Teflon) or Silicone** sleeving. Standard PVC will melt or become brittle near the 26cm tri-ply pan’s heat.

2. Load Cell to ADC Connector: GX12 Aviation Plug (4-Pin)

This is the connection between the pan's load cell and the electronics housing.

- **Type: GX12 (12mm) 4-Pin Metal Aviation Connector**.
- **Vibration Resistance:** These connectors feature a **threaded locking nut**. This is critical because the vibration from the **Stirrer Subsystem** would eventually wiggle a standard plastic connector loose.
- **Shielding:** The metal housing of the GX12 provides continuous shielding from the cable to the enclosure, preventing noise "leaks" at the junction point.
- **Analog(mV) Signal type** is transferred from Load Cell to ADC via this connector.

3. ADC to MCU Connector: JST-XH

Since the ADC (ADS1232) and MCU are typically housed on the same internal PCB or in the same enclosure:

- **Type: 2.54mm JST-XH (5-Pin)**.
- These are friction-locking headers that are compact but secure enough for consumer robotics. They ensure the data lines (Data, Clock, PDWN) stay connected during the robotic arm's high-speed path planning movements.
- ADC converts the signal from Analog to Digital and passes in serial form to Microcontroller via this connector.

Microcontroler Selection:-

For the **Posha** autonomous cooking setup, we would need a microcontroller (MCU) that can simultaneously handle **high-speed motor control** (for the robotic arm and stirrer), **high-precision sensor reading** (from the ADS1232), and **connectivity** for recipe updates.

1. Best Choice: ESP32-C3

The ESP32-S3 is the most balanced choice for consumer robotics due to its dual-core power and built-in wireless capabilities.

- **Dual-Core Processing:** we can dedicate one core entirely to the **robotic arm's path planning** and the other core to **weight data acquisition** from the ADS1232 and Wi-Fi communication. This prevents motor jitter from delaying your weight readings.
- **Advanced PWM:** It has 16 high-resolution PWM channels, allowing it to control the **stirrer motor** and the **multiple degrees of freedom (DOF)** in your robotic arm simultaneously.

- **AI Extensions:** The "S3" variant includes instructions for accelerating AI/math workloads, which is useful for complex trajectory calculations in autonomous cooking.
- **Built-in Wi-Fi/BLE:** Essential for a "smart" oven that needs to sync with a mobile app or download new recipes.

TASK 4:-

Embedded code to acquire data from ADC to MCU:-

```
#include "ADS1232.h"  // Professional library for the TI ADS1232
#define DOUT  18      // Data pin (connected to MCU MISO)
#define SCLK  19      // Clock pin (connected to MCU SCK)
#define PDWN  10      // Power Down pin (High = Active)
ADS1232 scale;
void setup() {
   Serial.begin(115200);
   // Initialize ADC with Pins
   scale.begin(DOUT, SCLK, PDWN);
   // Configuration for Posha Kitchen Environment
   scale.set_gain(ADS1232_GAIN_128); // Max sensitivity for small spice      doses
   scale.set_speed(ADS1232_SPEED_10); // 10SPS for 50/60Hz oven noise rejection
   scale.set_scale(2150.f);          // Calibration factor (grams per bit)
   scale.tare();                     // Zero out the 1.2kg pan weight
}
void loop() {
   // Read weight (averaging 10 samples to filter stirrer vibration)
   float weight = scale.get_units(10);
   Serial.print("Ingredient Weight: ");
   Serial.print(weight);
   Serial.println(" g");
   delay(500);
}
```

Task 5:-

Full PCB designing and Schematics shared in Google drive link.
PCB design drive link

Short briefing of how the System Works:-
Load cell -> GX12 Connector -> ADS1232 -> JST_XH -> ESP32-C3

1. The GX12 to ADS1232 Mapping:

GX12 Pin	Signal	ADS1232 Pin	Connection Detail/Logic
Pin 1	E+	16(REFP) & 18(AVDD)	Connects to your 5V Analog Power
Pin 2	E-	15(REFN)& 17(AGND)	Connects to your Analog Ground
Pin 3	S+	11(AINP1)	Place a 10ohm Resistor in series here
Pin 4	S-	12(AINN1)	Place a 10ohm Resistor in series here
Pin 5	Shield	Chassis Ground	Direct

2. ADS1232 To ESP32-S3 Mapping via JST_XH(5pin) :-

JST_XH Pin	Signal Name	ADS1232 Pin	ESP32-C3	Wave Function
Pin 1	VCC(3.3V)	1(DVDD)	2(VDD3.3V)	Digital Power Supply
Pin 2	GND	2,5,6(DGND)	33(GND)	Digital Ground Return
Pin 3	SCLK	23(SCLK)	26(GPIO19)	Wave1: Serial Clock
Pin 4	DOUT	24(DRDY/DOUT)	25(GPIO18)	Wave2: Data Out
Pin 5	PDWN	22(PWDN)	16(GPIO10)	Wave3: Power Control

Conclusion:-

My design of the **Posha Load Sensing Subsystem** successfully integrates **high-precision industrial** components within a challenging **thermal** and **mechanical** environment. By selecting a **5kg Stainless Steel Single-Point Load Cell** paired with the **TI ADS1232 24-bit ADC**, the system achieves the sensitivity required for milligram-level spice dosing while maintaining the robustness to handle a **1.0kg pan** and a **1.5kg payload**.

The implementation of **differential signalling**, **low-pass hardware filtering**, and **thermal isolation slots** on the PCB ensures that the electrical noise from the **stirrer motor** and the radiant heat from the **26cm tri-ply pan** do not compromise weight accuracy. Finally, the use of the **ESP32-C3** provides the computational power necessary for real-time robotic path planning and seamless smart-oven integration. This hardware-software synergy establishes a reliable foundation for fully autonomous household cooking.

Bill of Materials (BOM) - Posha Smart Oven Load Sensing Subsystem

Item	Qty	Reference	Description	Recommended Manufacturer/Part No.
1	1	**S1**	**Load Cell**: 5kg Single-Point, Stainless Steel, IP68, C3 Accuracy	**HBM PW15AH** or **Zemic L6E3**
2	1	**U1**	**ADC**: 24-Bit Analog-to-Digital Converter for Bridge Sensors	**Texas Instruments ADS1232IPW**
3	1	**U3**	**MCU**: ESP32-C3-WROOM-1 (Dual-Core, Wi-Fi/BLE)	**Espressif ESP32-C3-WROOM-1-N8**
4	1	**J1**	**Connector (Ext)**: 4-Pin GX12 Metal Aviation Plug (Panel Mount)	**Generic GX12-4P**
5	1	**J2**	**Connector (Int)**: 5-Pin JST-XH 2.50mm Vertical Header	**JST B5B-XH-A**
6	1	**U2**	**Voltage Ref**: 5V Low-Noise LDO (for Load Cell Excitation)	**TI LP2985-50** or **REF195**
7	2	**R1, R2**	**Resistor**: 10Ω, 1/8W, 1% Tolerance, 0805 SMD	**Panasonic ERJ-6GEYJ100V**

8	1	**C1**	**Capacitor (Diff)**: 100nF (0.1µF), 50V, X7R Ceramic, 0805 SMD	**Samsung CL21B104KBCNNNC**
9	1	**C2**	**Capacitor (Ref)**: 10µF, 16V, Tantalum (Polarized), Case A/B	**KEMET T491A106K016AT**
10	1	**Cable**	**Cable**: 4-Core Shielded Twisted Pair (Teflon/PTFE Jacket)	**Alpha Wire 58412**